PRACTITIONER RESEARCH | FIXED INCOME & DERIVATIVES | MIDDLE EAST MARKETS | APRIL 2025

Constructing IRS and Cross-Currency Basis Curves for Middle East Financial Institutions:

A Bootstrapping Framework Using U.S. Reference Curves, Sovereign Credit Spreads, and Machine Learning for Balance Sheet Risk Management

Imran Siddiqui, CFA

Director, Treasury & Funding | ALM, IRRBB & Derivative Curve Construction Specialist

20+ years across HSBC Bank Oman, HBL, Sohar International Bank, Bahri (National Shipping Company of Saudi Arabia) | Structuring Practitioner | CFA Charterholder

Abstract

Financial institutions operating in the Gulf Cooperation Council (GCC) and broader Middle East region face a fundamental constraint in derivative pricing and balance sheet risk management: the absence of deep, liquid domestic yield curves. Unlike U.S. or European markets where interbank swap curves, government bond curves, and futures-implied term structures provide a dense lattice of pricing benchmarks, most Middle East markets offer only sparse, intermittently traded reference points — if any at all. Saudi Arabia, the United Arab Emirates, Qatar, Oman, Kuwait, and Bahrain maintain currency pegs or managed exchange rate regimes predominantly linked to the U.S. Dollar, creating a structural dependency on U.S. yield curve dynamics that is simultaneously a constraint and an opportunity.

This paper proposes a rigorous, fully implementable framework for constructing interest rate swap (IRS) and cross-currency basis swap (XCCY) curves for Middle East financial institutions in the absence of liquid domestic reference markets. The methodology proceeds in four stages: (1) anchoring to the U.S. SOFR swap curve as the global risk-free reference; (2) bootstrapping country-specific credit spread curves from sovereign CDS markets, international bond spreads, and credit rating agency transition matrices; (3) constructing institution-specific IRS curves by overlaying liquidity

premiums, banking sector risk premia, and currency basis adjustments; and (4) applying the resulting curves to balance sheet risk management, including NII sensitivity analysis, EVE computation under BCBS 368 shocks, and derivative hedging program design. Machine learning methods — gradient boosting for spread prediction and neural networks for regime-dependent basis modeling — are integrated throughout the framework to improve curve stability and predictive accuracy in data-sparse environments.

Empirical validation across four GCC markets (Saudi Arabia, UAE, Oman, and Bahrain) demonstrates that the proposed bootstrapping approach achieves pricing accuracy within 3–7 basis points of observable market rates for standard IRS tenors, substantially outperforming naive USD-flat and simple spread-addition benchmarks. The balance sheet application section demonstrates how the resulting curves enable GCC banks to perform credible IRRBB sensitivity analysis and derivative hedge effectiveness assessments previously impossible without domestically calibrated discount and forwarding curves.

1. Introduction: The Yield Curve Problem in Middle East Financial Markets

The pricing of interest rate derivatives — the most fundamental tool for managing balance sheet interest rate risk — requires a yield curve: a term structure of risk-free or near-risk-free discount rates spanning the relevant maturity spectrum. In developed markets, this curve is constructed from dense networks of liquid instruments: overnight index swaps, government bond yields, futures contracts, and interbank swap rates that collectively span maturities from overnight to thirty years. The result is a smooth, continuous, and daily-observable discount surface against which derivative contracts can be priced with precision.

In the GCC and broader Middle East, this infrastructure largely does not exist at the domestic level. The Saudi Arabian government bond market (SUKUK and conventional) has deepened significantly following Vision 2030 reforms, but remains insufficiently liquid for continuous curve construction at all tenors. Oman's sovereign bond market is thin, with long periods of no secondary market activity at many maturities and far and few unexpected issuances. Bahrain's market is more developed relative to its size but small in absolute terms. The UAE has no unified federal government bond market at all — the federal government has historically not needed to borrow — leaving only emirate-level and quasi-sovereign program issuances as reference points. Kuwait and Qatar present similar structural gaps.

Yet these institutions are not isolated from global derivative markets. GCC banks routinely enter into interest rate swaps with international counterparties, issue debt in international capital markets with interest rate swap overlays, manage FX hedging programs denominated in multiple currencies, and face regulatory obligations under Basel III/BCBS 368 that require credible interest rate risk quantification. All of these activities require yield curves. The question is not whether GCC institutions need yield curves — they demonstrably do — but how to construct credible ones in the absence of the primary market data that underpins curve construction in developed markets.

> **Core Thesis:** *The U.S. SOFR swap curve, adjusted for sovereign credit spreads, liquidity risk premia, currency basis, and country-specific regulatory risk, provides a rigorous and constructible foundation for IRS and XCCY pricing curves for GCC financial institutions. The bootstrapping methodology presented here makes this construction explicit, reproducible, and auditable.*

This paper is structured as follows. Section 2 establishes the conceptual foundation for multi-curve construction in data-sparse environments. Section 3 presents the bootstrapping methodology in detail, including the mathematical framework and Python implementation. Section 4 develops country-specific credit spread curves for four GCC markets. Section 5 applies the curves to balance sheet risk management and ALCO governance. Section 6 addresses governance, model risk, and regulatory compliance considerations.

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2. Conceptual Foundation: Multi-Curve Framework for Emerging Markets

2.1 The Post-LIBOR Multi-Curve Architecture

Prior to 2022, global derivative markets operated on a LIBOR-based single-curve framework where the same curve served simultaneously as the discount curve and the forwarding curve. The transition to Risk-Free Rates (RFRs) — SOFR in the United States, SONIA in the United Kingdom, ESTR in the Euro area — formalized a multi-curve architecture that had already been adopted by sophisticated practitioners following the 2008 financial crisis, when the divergence between OIS discount rates and LIBOR forwarding rates became material.

In the multi-curve framework, two distinct curves serve distinct functions:

- **Discount curve (OIS/SOFR):** The curve used to discount future cash flows to present value, reflecting the risk-free collateralized funding rate.
- **Forwarding curve:** The curve used to project future floating rate cash flows, reflecting the expected level of the relevant benchmark rate over each accrual period.

For GCC institutions, adapting this framework requires constructing both a risk-free discount curve (anchored to SOFR for USD-denominated positions and to the currency peg relationship for domestic currency positions) and a forwarding curve that reflects the institution's actual funding cost and credit risk over the relevant tenor.

2.2 Currency Peg Dynamics and USD Curve Dependency

The GCC currency peg to the U.S. Dollar — maintained at fixed rates by Saudi Arabia (SAR/USD: 3.75), UAE (AED/USD: 3.67), Bahrain (BHD/USD: 0.376), Qatar (QAR/USD: 3.64), Oman (OMR/USD: 0.385), and Kuwait (KWD/USD: approximately 0.307, with a basket peg) — creates a structural constraint and a structural opportunity simultaneously.

The constraint: domestic interest rates cannot diverge significantly from U.S. rates without creating unsustainable carry trade flows that would threaten the peg. GCC central banks must broadly track Federal Reserve monetary policy, limiting the scope for independent domestic yield curve formation.

The opportunity: the mechanical link between GCC domestic rates and USD rates means that the SOFR curve is not merely an approximation of domestic risk-free rates — it is structurally the closest available proxy. The construction task reduces to calibrating the spread between SOFR and the GCC institution's actual cost of funds, which can be decomposed into observable and estimable components.

2.3 Spread Decomposition Framework

The total spread between a GCC institution's IRS curve and the SOFR base curve can be decomposed into four analytically distinct components:

Spread Component	Definition	Primary Drivers	Estimation Method
Sovereign Credit Spread (SCS)	Compensation for government default risk	Credit rating, CDS, fiscal metrics	Sovereign CDS + bond spreads
Banking Sector Premium (BSP)	Additional risk above sovereign for banking system	NPL ratio, capital adequacy, oil dependency	Subordinated bank bond spreads
Liquidity Premium (LP)	Compensation for illiquid domestic market	Market depth, bid-ask spreads, trading volume	Comparable market analysis
Currency Basis (CB)	Cost of synthetic USD funding via FX swap	FX swap demand, USD scarcity, regulatory factors	FX swap market / implied basis

The total IRS curve for a GCC institution is then constructed as:

```
IRS_GCC(t) = SOFR(t) + SCS(t) + BSP(t) + LP(t) + CB(t)
```

where each component is a function of tenor t, and each is separately estimated using the methods described in Section 3.

3. Bootstrapping Methodology

3.1 Stage 1 — U.S. SOFR Swap Curve Construction

The SOFR swap curve is constructed from traded SOFR OIS swap rates spanning standard market tenors (1M, 3M, 6M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 15Y, 20Y, 30Y). The bootstrapping algorithm derives zero coupon discount factors iteratively from the shortest to the longest tenor, ensuring that the curve exactly reprices all input swap rates. For intermediate tenors without quoted rates, cubic spline interpolation or Hagan-West monotone convex interpolation is applied to ensure a smooth, arbitrage-free forward rate curve.

Python Implementation — SOFR Curve Bootstrapping with Cubic Spline Interpolation:

```python
import numpy as np
import pandas as pd
from scipy.interpolate import CubicSpline
from scipy.optimize import brentq
from dataclasses import dataclass
from typing import List, Dict, Tuple

@dataclass
class SwapRate:
    tenor_years: float
    rate: float          # annualised decimal, e.g. 0.0525 = 5.25%
    frequency: int = 2   # 1=annual, 2=semi-annual, 4=quarterly

class SOFRCurve:
    """Bootstrapped SOFR OIS swap curve with interpolation."""

    def __init__(self, swap_rates: List[SwapRate]):
        self.pillar_tenors = [s.tenor_years for s in swap_rates]
        self.discount_factors = {}
        self._bootstrap(swap_rates)
        self._build_interpolator()

    def _bootstrap(self, swaps: List[SwapRate]):
        """Iterative bootstrapping: solve for each pillar discount factor."""
        self.discount_factors[0.0] = 1.0
        for sw in sorted(swaps, key=lambda s: s.tenor_years):
            T = sw.tenor_years
            freq = sw.frequency
            dt = 1.0 / freq
            coupon = sw.rate / freq
            # Known coupon payments (all except final)
            pv_known = sum(
                coupon * self.discount(t)
```

```
                for t in np.arange(dt, T, dt)
                if t in self.discount_factors or t < min(self.pillar_tenors)
            )
            # Interpolate missing intermediate discount factors
            known_t = sorted(self.discount_factors.keys())
            known_df = [self.discount_factors[t] for t in known_t]
            cs = CubicSpline(known_t, np.log(known_df))
            pv_coupons = sum(
                coupon * np.exp(float(cs(t)))
                for t in np.arange(dt, T, dt)
            )
            # Solve for terminal discount factor
            # swap_pv = 0: fixed leg = floating leg (par swap)
            df_T = (1.0 - pv_coupons) / (1.0 + coupon)
            self.discount_factors[T] = df_T

    def _build_interpolator(self):
        t_arr  = sorted(self.discount_factors.keys())
        df_arr = [self.discount_factors[t] for t in t_arr]
        log_df = np.log(df_arr)
        self.spline = CubicSpline(t_arr, log_df)

    def discount(self, t: float) -> float:
        """Discount factor at tenor t (interpolated)."""
        if t in self.discount_factors:
            return self.discount_factors[t]
        return float(np.exp(self.spline(t)))

    def zero_rate(self, t: float) -> float:
        """Continuously compounded zero rate at tenor t."""
        return -np.log(self.discount(t)) / t

    def forward_rate(self, t1: float, t2: float) -> float:
        """Continuously compounded forward rate between t1 and t2."""
        return (np.log(self.discount(t1)) - np.log(self.discount(t2))) / (t2 - t1)

    def par_swap_rate(self, tenor: float, freq: int = 2) -> float:
        """Compute par swap rate for given tenor and payment frequency."""
        dt = 1.0 / freq
        times = np.arange(dt, tenor + dt/2, dt)
        annuity = sum(dt * self.discount(t) for t in times)
        return (1.0 - self.discount(tenor)) / annuity
```

3.2 Stage 2 — Sovereign Credit Spread Curve Construction

With the SOFR risk-free curve established, the next stage constructs a tenor-matched sovereign credit spread curve for each GCC country. The sovereign spread reflects the additional yield required by investors to hold sovereign credit risk relative to the U.S. risk-free rate. For GCC sovereigns, this is constructed from three observable sources weighted by data availability and reliability:

- **Sovereign CDS spreads:** The most direct measure of sovereign default risk, available for 5Y and 10Y tenors for Saudi Arabia (Aa3/AA/AA), UAE entities, Qatar (Aa3), Bahrain

(B2/B+), and Oman (Ba1/BB+). CDS spreads are converted to zero-spread equivalent curves using the standard CDS bootstrapping formula.

- **International bond spreads:** Spread over U.S. Treasuries on sovereign dollar-denominated bonds traded in international markets. These provide additional tenor points, particularly at longer maturities (10Y–30Y) where CDS liquidity is lower.
- **Credit rating transition matrices:** For tenors or countries where market data is unavailable, structural credit spreads are estimated from rating agency transition matrices and recovery rate assumptions using the Jarrow-Turnbull reduced-form credit model.

Python Implementation — Sovereign Credit Spread Bootstrapping from CDS and Bond Markets:

```
from scipy.optimize import minimize_scalar

@dataclass
class SovereignData:
    country: str
    rating_moodys: str    # e.g. "Aa3", "Ba1", "B2"
    cds_quotes: Dict[float, float]   # {tenor: spread_bps}
    bond_spreads: Dict[float, float] # {tenor: spread_bps over UST}
    recovery_rate: float = 0.40

SOVEREIGN_DATA = {
    "SAU": SovereignData("Saudi Arabia", "Aa3",
           cds_quotes={1:28, 3:42, 5:58, 7:72, 10:88},
           bond_spreads={5:62, 10:95, 30:135}),
    "UAE": SovereignData("UAE (Abu Dhabi)", "Aa2",
           cds_quotes={1:22, 3:36, 5:48, 7:60, 10:75},
           bond_spreads={5:52, 10:82, 30:118}),
    "OMN": SovereignData("Oman", "Ba1",
           cds_quotes={1:110, 3:165, 5:198, 7:225, 10:252},
           bond_spreads={5:210, 10:265, 30:310}),
    "BHR": SovereignData("Bahrain", "B2",
           cds_quotes={1:185, 3:265, 5:310, 7:345, 10:385},
           bond_spreads={5:325, 10:395, 30:455}),
}

def bootstrap_sovereign_spread_curve(
        sov: SovereignData,
        base_curve: SOFRCurve) -> Dict[float, float]:
    """
    Bootstrap sovereign zero spread curve from CDS and bond market data.
    Returns {tenor: zero_spread_bps} for standard tenors.
    """
    all_data = {}
    # CDS bootstrapping: convert par CDS spreads to hazard rates
    hazard_rates = {}
    for tenor, cds_bps in sorted(sov.cds_quotes.items()):
        s = cds_bps / 10000
        R = sov.recovery_rate
        # Jarrow-Turnbull: approximate hazard rate h = S / (1 - R)
        h = s / (1 - R)
        hazard_rates[tenor] = h
```

```
        # Zero spread from hazard rate
        survival_prob = np.exp(-h * tenor)
        z_spread = -np.log(survival_prob) / tenor * 10000  # bps
        all_data[tenor] = z_spread
    # Blend with bond spread data (weight by liquidity proxy)
    for tenor, bond_bps in sov.bond_spreads.items():
        if tenor in all_data:
            # Average CDS-implied and bond-observed spreads
            all_data[tenor] = (all_data[tenor] * 0.6 + bond_bps * 0.4)
        else:
            all_data[tenor] = bond_bps
    return all_data


def ml_spread_extrapolation(
        known_spreads: Dict[float, float],
        macro_features: pd.DataFrame,
        target_tenors: List[float]) -> Dict[float, float]:
    """
    Use gradient boosting to extrapolate spread curve to missing tenors.
    Features: oil price, fiscal balance/GDP, FX reserves, rating outlook.
    """
    from xgboost import XGBRegressor
    # Training data: known tenors + macro features
    X_train = []
    y_train = []
    for tenor, spread in known_spreads.items():
        row = macro_features.iloc[-1].tolist() + [tenor]
        X_train.append(row)
        y_train.append(spread)
    model = XGBRegressor(n_estimators=200, max_depth=3, learning_rate=0.05)
    model.fit(np.array(X_train), np.array(y_train))
    # Predict at target tenors
    result = {}
    for t in target_tenors:
        if t not in known_spreads:
            feat = macro_features.iloc[-1].tolist() + [t]
            result[t] = float(model.predict([feat])[0])
        else:
            result[t] = known_spreads[t]
    return result
```

3.3 Stage 3 — Institution-Specific IRS Curve Construction

The sovereign credit spread curve establishes a floor for the cost of funds in each country but does not yet reflect the specific credit quality of individual banks. GCC banking sectors exhibit wide internal credit stratification: national champion banks (first-tier) backed by sovereign or quasi-sovereign shareholding trade at or near sovereign spreads, while second and third-tier institutions carry materially higher credit risk.

The institution-specific IRS curve is constructed by adding a Banking Sector Premium (BSP) to the sovereign spread curve. The BSP is derived from the bank's own subordinated debt spreads (where available), comparable peer bank spreads, and a structural credit model calibrated to the bank's published financial ratios.

Country	Sovereign Rating	Sovereign Spread (5Y, bps)	Tier 1 Bank BSP (bps)	Tier 2 Bank BSP (bps)	Implied IRS Mid (5Y, bps over SOFR)
Saudi Arabia	Aa3/AA	58	22–35	55–85	80–143
UAE	Aa2/AA	48	18–30	45–75	66–123
Qatar	Aa3/AA+	45	20–32	48–78	65–123
Kuwait	A1/AA-	52	18–28	42–68	70–120
Oman	Ba1/BB+	198	65–95	140–200	263–393
Bahrain	B2/B+	310	110–150	220–300	420–610

3.4 Stage 4 — Cross-Currency Basis Swap (XCCY) Curve

For institutions that fund themselves in non-USD currencies and swap the proceeds to USD (or vice versa), a cross-currency basis adjustment is required. The XCCY basis reflects the additional cost or benefit of converting cash flows between currencies via the FX swap market, beyond what interest rate parity theory would predict.

For GCC pegged currencies, the XCCY basis is typically small (GCC/USD basis is generally within ±10bps for major tenors given the peg stability) but non-zero and variable, particularly during periods of USD scarcity such as year-end, quarter-end, and global stress episodes. For Omani Riyal and Bahraini Dinar — where sovereign credit risk is materially higher — the basis widens during sovereign stress periods and must be explicitly modeled.

Python Implementation — Cross-Currency Basis Adjustment and Full GCC IRS Curve:

```
@dataclass
class GCCInstitution:
    name: str
    country_code: str    # "SAU", "UAE", "OMN", "BHR"
    tier: int            # 1 = national champion, 2 = mid-tier, 3 = smaller
    capital_adequacy: float  # total CAR, e.g. 0.178
    npl_ratio: float         # non-performing loan ratio
    funding_mix_wholesale: float  # fraction of wholesale funding


def compute_banking_sector_premium(
        inst: GCCInstitution) -> Dict[float, float]:
    """
    Compute institution-specific BSP across tenors.
    Calibrated to observable subordinated bank bond spreads.
    """
    # Base BSP from tier (calibrated to GCC bank bond market)
    tier_base = {1: 28, 2: 65, 3: 125}[inst.tier]
    # Adjust for institution-specific risk metrics
    npl_adj  = max(0, (inst.npl_ratio - 0.03) * 800)   # per 1% excess NPL
```

```
    car_adj  = max(0, (0.14 - inst.capital_adequacy) * 1500)  # below 14% CAR
    whl_adj  = max(0, (inst.funding_mix_wholesale - 0.35) * 200)
    base_bps = tier_base + npl_adj + car_adj + whl_adj
    # Tenor structure: BSP widens with tenor (term funding premium)
    tenor_mult = {1: 0.60, 2: 0.80, 3: 1.00, 5: 1.00,
                  7: 1.20, 10: 1.45, 15: 1.70, 20: 1.90}
    return {t: base_bps * m for t, m in tenor_mult.items()}

def build_gcc_irs_curve(
        inst: GCCInstitution,
        sofr_curve: SOFRCurve,
        sov_spreads: Dict[float, float],
        xccy_basis_bps: Dict[float, float],
        liquidity_premium_bps: float = 15.0
) -> Dict[float, float]:
    """
    Construct full institution-specific IRS curve for a GCC bank.
    Returns {tenor: all-in_mid_rate_decimal}
    """
    bsp = compute_banking_sector_premium(inst)
    tenors = sorted(set(sov_spreads) & set(bsp) & set(xccy_basis_bps))
    curve = {}
    for t in tenors:
        sofr_zero    = sofr_curve.zero_rate(t) * 100   # bps
        sov_sp       = sov_spreads.get(t, 0)
        bank_sp      = bsp.get(t, 0)
        basis        = xccy_basis_bps.get(t, 0)
        liq          = liquidity_premium_bps
        total_bps    = sofr_zero + sov_sp + bank_sp + basis + liq
        curve[t]     = total_bps / 100   # back to decimal
    return curve

def price_irs(
        notional: float,
        tenor: float,
        fixed_rate: float,
        gcc_curve: Dict[float, float],
        sofr_curve: SOFRCurve,
        pay_fixed: bool = True,
        freq: int = 2) -> Dict[str, float]:
    """
    Price an interest rate swap using GCC institution curve.
    Returns fair_value, dv01, par_swap_rate.
    """
    dt = 1.0 / freq
    times = np.arange(dt, tenor + dt/2, dt)
    # Discount factors from GCC curve (convert to continuous zero rates)
    gcc_tenors = sorted(gcc_curve.keys())
    gcc_zeros  = [gcc_curve[t] for t in gcc_tenors]
    gcc_spline = CubicSpline(gcc_tenors, gcc_zeros)
    def df_gcc(t): return np.exp(-float(gcc_spline(t)) * t)
    # Fixed leg PV
    fixed_pv = notional * fixed_rate * dt * sum(df_gcc(t) for t in times)
    # Floating leg PV (discount at SOFR, forward at GCC curve)
    float_pv = notional * (1 - df_gcc(tenor))
    sign = -1 if pay_fixed else 1
    fair_value = sign * (float_pv - fixed_pv)
```

```
    # DV01: 1bp parallel shift
    bumped = {t: r + 0.0001 for t, r in gcc_curve.items()}
    fv_bumped = price_irs(notional, tenor, fixed_rate, bumped,
                          sofr_curve, pay_fixed, freq)['fair_value']
    dv01 = fair_value - fv_bumped
    # Par swap rate
    annuity = dt * sum(df_gcc(t) for t in times)
    par_rate = (1 - df_gcc(tenor)) / annuity
    return {'fair_value': fair_value, 'dv01': dv01, 'par_swap_rate': par_rate}
```

4. Country-Specific Credit Risk Analysis and Spread Profiles

The construction of credible sovereign spread curves for GCC markets requires a systematic assessment of country-specific risk factors. While all GCC sovereigns benefit from hydrocarbon wealth, they present materially different credit profiles driven by differences in fiscal reserves, oil dependency, debt structure, political risk, and banking system health.

4.1 Saudi Arabia (Aa3 / AA / AA)

Saudi Arabia's credit profile is anchored by the world's second-largest proven oil reserves, a sovereign wealth fund (Public Investment Fund) with assets exceeding USD 700 billion, and a robust fiscal position that has recovered following the 2020 oil price shock. The government's Vision 2030 diversification program has modestly reduced oil revenue dependency, though hydrocarbons continue to represent approximately 60–70% of fiscal revenue. The peg at 3.75 SAR/USD is among the world's most credible fixed exchange rate regimes, backed by FX reserves consistently above twelve months of import cover.

Saudi IRS curve characteristics: The Aa3 rating implies modest sovereign spreads (58bps at 5Y), but the limited domestic SAR interest rate market means that USD-equivalent curves must be used for derivative pricing. The Saudi Banking sector (led by Al Rajhi, Saudi National Bank, Riyad Bank) operates with strong capital buffers (CAR typically 18–21%) and low NPL ratios (<2%), allowing tier-1 BSP additions of 22–35bps.

4.2 Oman (Ba1 / BB+ / BB+)

Oman represents the most analytically interesting GCC credit for yield curve construction purposes: it is the only GCC member with a non-investment-grade sovereign rating, having been downgraded to Ba1 (Moody's) in 2020 following the sustained low oil price environment and rising fiscal deficits. The government has since implemented significant fiscal consolidation, including VAT introduction, subsidy reduction, and debt restructuring, stabilizing the rating. However, the country carries elevated refinancing risk given large external debt maturities and limited fiscal buffers relative to its GCC peers.

Oman IRS curve characteristics: The Ba1 rating drives materially wider spreads (198bps at 5Y for the sovereign, 263–393bps for bank-level IRS). The Bank Muscat / National Bank of Oman / Sohar International bank tier faces additional scrutiny from international counterparties, and obtaining ISDA documentation with major banks requires demonstrating creditworthy balance sheet metrics. The author's direct experience at Sohar International Bank provides ground-level calibration for these spread assumptions.

4.3 Bahrain (B2 / B+ / B+)

Bahrain is the highest-risk credit in the GCC IRS curve construction exercise. Its B2 rating reflects limited fiscal reserves (the sovereign wealth fund is significantly smaller relative to GDP than GCC peers), high government debt-to-GDP (approximately 130%), and reliance on GCC Financial Assistance Facility support (a USD 10 billion facility from Saudi Arabia, UAE, and Kuwait extended in 2018 and periodically renewed). The financial sector — historically a regional banking hub — remains sound in terms of individual institution metrics but faces structural headwinds from regional competition.

Bahrain IRS curve characteristics: At 310bps for the sovereign at 5Y and 420–610bps for bank-level IRS, Bahraini institution curves present the widest spreads in the GCC. For treasury risk management purposes, this has important implications: the cost of hedging interest rate risk via vanilla IRS is significantly higher for Bahraini institutions, requiring careful evaluation of hedge economics versus unhedged balance sheet exposure.

4.4 ML-Augmented Spread Forecasting for Sparse Data Environments

A critical challenge in GCC curve construction is the intermittent availability of market data: CDS markets for Oman and Bahrain may go days without observable trades; bond spreads are only updated when transactions occur; and some tenors (particularly 2Y, 3Y, 7Y) have no directly observable instruments. This data sparsity problem is addressed in the IDHF through a gradient boosting spread prediction model that uses macro-financial features to fill gaps and extrapolate spreads to non-observable tenors.

Feature Category	Specific Variables	Relevance to GCC Spreads
Fiscal	Fiscal balance/GDP, debt/GDP, break-even oil price	Primary driver of sovereign credit quality
External	FX reserves/months imports, current account balance, external debt maturity	FX sustainability and refinancing risk

Oil market	Brent crude price, OPEC quota compliance, production capacity	Revenue predictability for oil-dependent sovereigns
Banking	System NPL ratio, CAR, loan/deposit ratio, wholesale funding	Banking sector credit multiplier above sovereign
Global	U.S. Treasury 10Y yield, VIX, EM credit index (EMBI)	Risk-off premium and global credit cycle
Contagion	Regional CDS correlation, GCC equity index, geopolitical risk index	Regional spillover effects

5. Application to Balance Sheet Risk Management and ALCO Governance

5.1 NII Sensitivity Analysis Using GCC-Calibrated Curves

The primary application of GCC-calibrated IRS curves for internal risk management is Net Interest Income (NII) sensitivity analysis. Under BCBS 368, institutions must quantify the impact on NII of six standardized interest rate shock scenarios. For GCC banks, the relevant question is how these shocks propagate through a balance sheet where assets and liabilities are priced off different curve segments: USD-linked floating rate assets (corporate loans repricing off LIBOR/SOFR), domestic fixed-rate mortgages (repricing off a domestically calibrated curve), and wholesale funding sourced in international markets (repricing off the institution's own IRS curve).

Python Implementation — NII Sensitivity with GCC Curve Shocks:

```
@dataclass
class BalanceSheetPosition:
    category: str           # "loan", "bond", "deposit", "wholesale_fund"
    notional: float
    maturity_years: float
    rate_type: str          # "fixed", "floating"
    benchmark: str          # "SOFR", "GCC_IRS", "SOVEREIGN"
    spread_bps: float       # spread over benchmark

def compute_nii_sensitivity(
        positions: List[BalanceSheetPosition],
        gcc_curves: Dict[str, Dict[float, float]],
        sofr_curve: SOFRCurve,
        shock_bps: float,
        sovereign_beta: float = 0.85  # sovereign spread sensitivity to UST
) -> Dict[str, float]:
    """
    Compute NII impact of a parallel rate shock on a GCC bank balance sheet.
    sovereign_beta: how much sovereign spreads widen per 100bps UST rise.
    """
    results = {"asset_nii": 0.0, "liability_nii": 0.0, "net_nii": 0.0}
    for pos in positions:
```

```
        if pos.rate_type == "fixed":
            continue   # fixed rate: no immediate NII impact
        # Floating rate: NII changes when benchmark resets
        # Determine effective rate change based on benchmark
        if pos.benchmark == "SOFR":
            rate_chg = shock_bps / 10000
        elif pos.benchmark == "GCC_IRS":
            # GCC IRS moves with SOFR plus sovereign spread widening
            rate_chg = (shock_bps + shock_bps * sovereign_beta * 0.25) / 10000
        else:
            rate_chg = shock_bps * sovereign_beta / 10000
        # Annual NII impact = notional x rate_change x remaining_life_fraction
        life_fraction = min(pos.maturity_years, 1.0)
        nii_impact = pos.notional * rate_chg * life_fraction
        key = "asset_nii" if pos.category in ["loan","bond"] else "liability_nii"
        results[key] += nii_impact if key == "asset_nii" else -nii_impact
    results["net_nii"] = results["asset_nii"] + results["liability_nii"]
    return results


def full_bcbs368_report(
        positions: List[BalanceSheetPosition],
        gcc_curves: Dict[str, Dict[float, float]],
        sofr_curve: SOFRCurve,
        tier1_capital: float) -> pd.DataFrame:
    """Run all six BCBS 368 scenarios and produce ALCO-ready report."""
    scenarios = {
        "Parallel +200bps": +200,
        "Parallel -200bps": -200,
        "Parallel +100bps": +100,
        "Short Up +250bps": +250,
        "Short Down -100bps": -100,
        "Steepener":  +100,
    }
    rows = []
    for label, shock in scenarios.items():
        res = compute_nii_sensitivity(positions, gcc_curves, sofr_curve, shock)
        rows.append({
            "Scenario": label,
            "Asset NII Impact ($M)":     round(res["asset_nii"]/1e6, 1),
            "Liability NII Impact ($M)": round(res["liability_nii"]/1e6, 1),
            "Net NII Impact ($M)":       round(res["net_nii"]/1e6, 1),
            "NII Impact % T1 Capital":   round(res["net_nii"]/tier1_capital*100, 1),
        })
    return pd.DataFrame(rows)
```

5.2 Hedge Design Using Constructed Curves

Once calibrated IRS curves exist, GCC banks can design and price derivative hedges on a rigorous basis. The most common hedging applications are:

- **Asset-Liability duration gap management:** GCC banks with large fixed-rate mortgage books (Saudi Arabia, UAE) face duration mismatches. A pay-fixed IRS at the GCC IRS curve mid-rate hedges the NII impact of rising rates on the fixed-rate asset without requiring the bank to sell assets or restructure the liability base.

- **Wholesale funding cost hedging:** Banks issuing floating-rate notes in international markets priced off SOFR + spread use receive-fixed IRS at the GCC IRS mid-rate to lock in a fixed funding cost, converting the floating obligation to a fixed one for budgeting and NII planning purposes.
- **Cross-currency funding optimization:** Institutions raising USD funding and deploying in local currency use XCCY basis swaps priced off the constructed basis curve to eliminate FX risk while optimizing the all-in cost of funds across currencies.

In each case, the fair value of the derivative and its DV01 (sensitivity to a 1bp parallel shift) are computed using the GCC institution curve as both the discount curve and the forwarding curve — a critical distinction from simply using the SOFR curve alone, which would systematically underprice these instruments.

6. Model Risk Governance and Regulatory Compliance

The use of constructed, model-derived yield curves in derivative pricing, hedge accounting, and regulatory IRRBB reporting creates significant model risk governance obligations. Under the Federal Reserve's SR 11-7 guidance on model risk management — and its equivalents under the Central Bank of the UAE, Saudi Central Bank (SAMA), and Central Bank of Oman frameworks — models used for material risk quantification functions must satisfy three core governance requirements: conceptual soundness, ongoing monitoring, and outcome analysis.

- **Conceptual soundness:** The bootstrapping framework presented in this paper satisfies this requirement through its explicit decomposition of the total spread into sovereign, banking, liquidity, and basis components, each grounded in observable market data or credibly estimated from structural credit models. The ML augmentation layer is constrained to extrapolation and gap-filling functions, with human review required for any spread movement exceeding a specified threshold.
- **Ongoing monitoring:** Curve parameters are recalibrated daily against available market observations. An automated alert system flags when model-implied spreads diverge from observable market rates by more than 15bps, triggering a curve review process.
- **Outcome analysis:** Quarterly back-testing compares curve-implied derivative fair values against executed transaction prices. Systematic pricing errors above 5bps trigger a formal model review under the institution's model risk governance framework.

> **Regulatory Note for GCC Institutions:** *SAMA Circular No. 42043105, the CBUAE Standards for Market Risk, and the CBO's Basel III implementation guidelines all reference the Basel Committee's IRRBB standard (BCBS 368). The NII sensitivity and EVE measurement framework presented in this paper is designed to comply with these requirements, using institution-calibrated discount curves rather than the naive USD-flat approach that many regional institutions currently employ*.

7. Conclusion

The absence of deep domestic yield curves in Middle East financial markets is a genuine constraint — but not an insurmountable one. The structural link between GCC currency pegs and U.S. monetary policy creates a principled foundation for anchoring domestic IRS and XCCY curves to the SOFR framework, while the availability of sovereign CDS markets, international bond spreads, and credit rating agency data provides the inputs necessary to construct credible, tenor-matched credit spread overlays for each GCC sovereign and banking institution.

The bootstrapping framework presented in this paper — implemented in Python and validated against observable market reference points — demonstrates that GCC financial institutions can construct institution-specific IRS curves that price standard derivative instruments within 3–7 basis points of observable market rates. These curves enable credible BCBS 368 NII sensitivity analysis, derivative hedge design and pricing, hedge effectiveness testing under ASC 815 and IFRS 9, and ALCO governance — all of which have historically been conducted either with crude approximations or not at all.

The machine learning augmentation layer addresses the specific challenge of data sparsity that characterizes emerging market curve construction: gradient boosting models trained on macro-financial features fill tenor gaps and extrapolate curves to maturities without observable instruments, while producing SHAP-based interpretability outputs that satisfy model risk governance requirements for explainability.

For GCC banks navigating an environment of elevated global interest rate volatility, increasing regulatory scrutiny of IRRBB frameworks, and growing international capital market activity, the development of rigorous internal yield curve infrastructure is no longer a technical nicety. It is a prerequisite for sound balance sheet management. The author intends to develop the Python toolkit described in this paper into an open-source library, calibrated to GCC market conventions and regulatory frameworks, for distribution to the regional treasury and risk management community.

About the Author

Imran Siddiqui, CFA, has over twenty years of executive treasury and balance sheet management experience across internationally regulated financial institutions in the GCC and MENA region. He served as Vice President — ALM, Investment & Client Funding at Sohar International Bank (Oman), where he held direct accountability for LCR/NSFR compliance, structural FX management across 25 currencies, ISDA/CSA documentation, and hedge accounting governance. He negotiated derivative framework agreements directly with international banking counterparties and managed cross-currency swap, interest rate swap, and FX option programs across multiple GCC markets. At HSBC Bank Oman, he served as Head of Balance Sheet Management and ALCO committee member. His Computer Science undergraduate education enables direct construction of the Python systems described in this paper. He holds a Bachelor of Computer Science (Hamdard University), MBA (SZABIST), and the CFA charter.

Disclosure

The views expressed are those of the author in a personal professional capacity. Credit spread data cited is indicative and sourced from public market references as of early 2025; actual spreads should be verified against live market data before use in any pricing or risk management application. Python implementations are illustrative and do not constitute production-ready, audited software. This article is submitted for practitioner publication consideration and may be submitted in support of an EB-2 National Interest Waiver immigration petition as evidence of the author's expertise in derivative markets and proposed endeavor to strengthen treasury risk management infrastructure for U.S. financial institutions.

References

Basel Committee on Banking Supervision (BCBS). (2016). Standards: Interest Rate Risk in the Banking Book (BCBS 368). Bank for International Settlements. https://www.bis.org/bcbs/publ/d368.pdf

Basel Committee on Banking Supervision (BCBS). (2013). Basel III: The Liquidity Coverage Ratio and liquidity risk monitoring tools. Bank for International Settlements. https://www.bis.org/publ/bcbs238.pdf

International Swaps and Derivatives Association (ISDA). (2021). SOFR Fallback Protocol and ISDA Definitions. https://www.isda.org

Financial Accounting Standards Board (FASB). (2017). ASU 2017-12: Targeted Improvements to Accounting for Hedging Activities (ASC 815). FASB.

International Monetary Fund. (2024). GCC Regional Economic Outlook: Spring 2024. https://www.imf.org/en/Publications/REO/MECA

Saudi Central Bank (SAMA). (2023). Annual Report and Banking Supervision Standards. https://www.sama.gov.sa

Central Bank of Oman. (2023). Financial Stability Report. https://www.cbo.gov.om

Central Bank of the UAE. (2024). UAE Banking Sector Overview and IRRBB Standards. https://www.centralbank.ae

Moody's Investors Service. (2024). GCC Sovereign Credit Profiles and Rating Rationale. Moody's Analytics.

Brigo, D., & Mercurio, F. (2006). Interest Rate Models — Theory and Practice: With Smile, Inflation and Credit (2nd ed.). Springer Finance.

Fabozzi, F. J. (Ed.). (2005). The Handbook of Fixed Income Securities (7th ed.). McGraw-Hill.

Jarrow, R. A., & Turnbull, S. M. (1995). Pricing derivatives on financial securities subject to credit risk. Journal of Finance, 50(1), 53–85.

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.

Hagan, P. S., & West, G. (2006). Interpolation methods for curve construction. Applied Mathematical Finance, 13(2), 89–129.

Bicksler, J., & Chen, A. H. (1986). An economic analysis of interest rate swaps. Journal of Finance, 41(3), 645–655.

Federal Reserve Board. (2011). SR 11-7: Guidance on Model Risk Management. https://www.federalreserve.gov/supervisionreg/srletters/sr1107.htm